

Jaekyung Jackie Lee

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RESEARCH SUMMARY

Ph.D. candidate specializing in autonomous systems, with research in multi-agent path planning, optimization, and dynamics and control of unmanned aerial and ground vehicles. Experienced in bridging algorithmic research with hardware deployment through ROS2, high-fidelity simulation, and sim-to-real transfer.

EDUCATION

Texas A&M University (Advisor: Dr. Sivakumar Rathinam)

College Station, TX

Ph.D. in Mechanical Engineering, Autonomous Systems Lab

Aug. 2022 – May 2027 (expected)

- Research Focus: Scalable optimization algorithms for multi-agent systems, distributed decision-making, and simulation-based performance analysis.

Kyungwoon University

South Korea

B.S. in Unmanned Aerial and Autonomous Vehicle Engineering, GPA: 3.9/4.0

Mar. 2018 – Feb. 2022

- Awards: *Summa Cum Laude*; Merit-based Scholarship with Highest Honors (three semesters); Full-tuition Scholarship (four years); National Science and Engineering Scholarship, Ministry of Education.

PUBLICATIONS

- **Lee, J.J.**, Rathinam, S. “Collaborative UAV-UGV Area Coverage under Battery Constraints: Team Orienteering Path Planning and Optimization-Based Scheduling.” *International Conference on Unmanned Aircraft Systems (ICUAS)*, May. 2025.
- Kumar, N., **Lee, J.J.**, Rathinam, S., Darbha, S., Sujit, P.B., & Raman, R. “The Persistent Robot Charging Problem for Long-Duration Autonomy.” *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2191–2198, Mar. 2025.
- **Lee, J.J.**, Rathinam, S. “A Meta-Heuristic Approach for an Aerial-Ground Vehicle Path Planning Problem.” *AIAA SCITECH Forum*, Jan. 2024.

RESEARCH & INDUSTRY EXPERIENCE

Magna Electronics Inc. | Advanced Driver Assistance Systems Engineer

Lowell, MA

ADAS Engineer Intern (Advisor: Ashesh Goswami)

May 2026 – Aug 2026

- Calibrated the perception front end of an automated **valet parking** system by computing **fish-eye camera intrinsics** and **camera-IMU extrinsics** to anchor downstream localization, free-space estimation, and closed-loop parking control.
- Brought up **GMSL2/FAKRA automotive cameras on NVIDIA Jetson AGX Orin**, modifying and rebuilding the vendor camera driver for the team's target JetPack/L4T configuration, and authored the bring-up documentation the team now works from.
- Containerized the perception development environment in **Docker with ROS2** to give the team a reproducible, hardware-in-the-loop build shared across developers.
- Developed collaboratively through **Git pull requests and peer code review** in a multi-developer production codebase, and validated results on bench and in-vehicle setups rather than in simulation alone.

Texas A&M University, Autonomous Systems Lab

College Station, TX

Graduate Research Assistant (Advisor: Dr. Sivakumar Rathinam)

Aug. 2022 – Present

- Developed a UAV trajectory optimization method incorporating camera aspect ratio to minimize overlapping coverage, achieving **10–20%** energy and path efficiency gains over conventional grid-based methods, with improvements up to **30%** in obstacle-rich environments.
- Engineered a **distributed optimization framework** for multi-agent path planning, reducing computational latency by **32%** through a real-time adaptive algorithm.
- Implemented a **Team Orienteering Problem (TOP)** heuristic solver in Gurobi, optimizing task allocation across 3K+ geo-coordinates for a custom UAV and a Clearpath UGV equipped with a manipulator.
- Integrated depth cameras, LiDAR, radio, and magnetic-field sensors into a Python-based simulation platform to evaluate autonomous vehicle performance under dynamic constraints.

Amazon Robotics | Warehouse Mobile Robot R&D

Applied Scientist II Intern (Advisor: Dr. Yuri Ivanov)

North Reading, MA

May 2025 – Dec. 2025

- Built a high-fidelity ROS2 + Isaac Sim simulation environment for large-scale evaluation of perception and closed-loop PID control pipelines, enabling robust testing under sensor noise, dynamic disturbances, and edge-case scenarios.
- Designed and demonstrated closed-loop motion and state-estimation (EKF) pipelines, integrating LiDAR-based perception with real-time trajectory execution, and executed sim-to-real transfer onto embedded platforms (Raspberry Pi, CAN bus, TI board, ESP32).
- Packaged research prototypes into reusable ROS2 modules and deployable robotic workflows, establishing end-to-end simulation-to-hardware validation pipelines for long-term system integration.

DEVCOM Army Research Lab | Collaborative Project

Graduate Researcher (Advisors: Dr. Sivakumar Rathinam, Dr. James Humann)

College Station, TX

Aug. 2022 – Jul. 2024

- Built a Python simulation system for NP-hard heuristic path planning, coordinating 4 heterogeneous aerial and ground vehicles over a **72-hour** persistent surveillance and task-solving mission.
- Formulated optimization-based motion strategies for heterogeneous robotic systems with physical and battery constraints, bridging planning and execution layers.
- Partnered with UGV hardware teams to validate planning algorithms through integrated simulation and field tests.

Kyungwoon University, Autonomous & Intelligent Robotics Lab

Project Leader — Independent Project: PAV Tri-eVTOL & Launch-Type Foldable Drone

South Korea

Sep. 2020 – Nov. 2021

- Led a team to design a tiltrotor eVTOL control system using PX4 autopilot (C++), optimizing motor efficiency through real-time convex optimization.
- Engineered a launch-type foldable drone mortar mechanism and control system on Pixhawk (C), and fabricated a hand-layered carbon-composite airframe with corresponding 3D CAD model.
- Prototyped a Raspberry Pi-based drone control system with MAVProxy, improving structural dynamics and control efficiency.
- Won \$13K at the 2021 International PAV Technology Contest for a scalable UAV simulation model.

TEACHING EXPERIENCE

MEEN 432: Automotive Engineering

Sep. 2026 – present

- Deliver hands-on instruction on automotive vehicle systems, guiding students with MATLAB and Simulink.
- Support **60+** students on experimental setup, data acquisition, and troubleshooting to ensure accurate measurement and recording by clarifying concepts and answering questions in dynamic control systems.

MEEN 365: Dynamic Systems and Control Lab

Jan. 2025 – May 2026

- Deliver hands-on instruction on AC/DC motor and slider-pendulum system measurement, guiding students through calibration with MATLAB and Simulink.
- Mentor **80+** students on experimental setup, data acquisition, and troubleshooting to ensure accurate measurement and recording.
- Support **540** students by clarifying concepts and answering questions in dynamic control systems.

MEEN 225: Engineering Mechanics (Honors)

Aug. 2024 – Dec. 2024

- Facilitated recitation sessions to reinforce lecture material and deepen student understanding.
- Supported **250+** students by clarifying concepts and answering questions in engineering mechanics.

AWARDS & HONORS

- **Korea-US Science Cooperation Center (KUSCO) Graduate Scholarship** - \$2,000 (2025)
- **Susan M. Arseven '75 'Make-a-Difference' Memorial Award**, Women in Science and Engineering (WISE) - \$1,000 (2025)
- **Summer Research Grant by J. Mike Walker '66 Department of Mechanical Engineering (TAMU)** - \$7,875 (2024)
- **Second Place, 2024 US-Korea Conference (UKC), Atlanta - Poster Award** for "A Meta-heuristic Approach Algorithm for an Unmanned Aerial-Ground Vehicles Path Planning Platform," J. Lee, S. Rathinam (Aug. 2024)

- **Hanwha Aerospace CEO Award**, 3rd National University Student Capstone Contest - Team Leader, “*Development of Launch-Type Foldable Quadcopter*,” Society for Aerospace System Engineering, Gyeongju, Korea (Nov. 2021)
- **Second Place, 2021 International PAV Technology Contest** - \$13,000 for downsized Uber-Hyundai S-A1 PAV eVTOL (12-motor) dynamics and control design (2021)
- **2019 Rocket Launch Competition**, National Universities Rocket Association (NURA) - Designed an Arduino- and servomotor-based parachute ejection system responsive to rocket posture changes (2019)
- **Third Place, 2019 Korea IT Businesswomen’s Association (IBWA) ICT Mentoring** - Team Leader, deep-learning-based hidden-camera detection technology, mobile app, and website (2019)
- **Second Place, 2018 Korea IT Businesswomen’s Association (IBWA) ICT Mentoring** - Drone-ICT Convergence solar power plant panel smart diagnosis technology (2018)

LEADERSHIP EXPERIENCE

Korean-American Scientists and Engineers Association (KSEA)

Aug. 2022 – Aug. 2025

President (Texas A&M Chapter) & National Secretary, UKC 2025

- Served as National Secretary for the 2025 US-Korea Conference (UKC), coordinating nationwide planning teams and organizing technical workshops for **1,500+** attendees and **200+** industry and academic speakers.
- Led the Texas A&M chapter for one year, orchestrating multi-institutional events and a career development forum featuring university faculty.

Mechanical Engineering Female Graduate Students (MEFEGs)

Jan. 2024 – Jan. 2025

Officer

- Organized departmental events to foster a stronger graduate student community.
- Managed the official club website to improve communication and member engagement.

Kyungwoon University Club Grin-Narae

Mar. 2019 – Dec. 2020

Founder and President

- Founded a drone development club focused on modular solutions for land, sea, and aerial applications.
- Secured a 2020 Prospective Start-up Package grant from the Korea Institute of Startup & Entrepreneurship Development for “*Non-face-to-face Emergency Rescue PAVs*.”
- Won 2020 Designated University Startup Club funding from the Korea Expressway Corporation, enabling lightweight quadcopter production using DJI Inspire 2 components.

TECHNICAL SKILLS

Programming: C++, Python, C, MATLAB, Simulink, Julia

Tools & Frameworks: ROS2, Docker, Isaac Sim, Gurobi, Linux, OpenCV, Kalibr, CATIA, Git

Core Expertise: Autonomous Systems, Multi-Agent Path Planning, Optimization, Sim-to-Real Transfer, Vehicle Dynamics, PID Control, Numerical & Statistical Methods

Hands-on: LiDAR/Camera/IMU Integration, Camera Intrinsic & Extrinsic Calibration, Jetson/GMSL2 Camera Bring-up, UAV & UGV Prototyping, Soldering, Mechanical Assembly

CERTIFICATIONS

- **FAA Part 107 Certified Remote Pilot** (Oct. 2022)
- **Ultralight Vehicle Pilot**, Korea Transportation Safety Authority (Jun. 2020)